BUILD WITH AI: CODE FOR COMMUNITIES · TRACK 5 · TEAM NEBULA NOMADS

CycloneSentinel

AI-powered cyclone impact & infrastructure vulnerability forecaster: from **satellite + real-time weather** to **Gemini-reasoned, auto-dispatched early warnings** before landfall.

Gemini 3.7 Flash | Google Earth Engine | Cloud Run | NASA VIIRS · GPM IMERG | Open-Meteo real-time | CAP 1.2 / SACHET

Live: blop77.github.io/cyclone-sentinel

github.com/Blop77/cyclone-sentinel

THE PROBLEM

Warnings say *where* the storm goes. Responders need to know *what breaks*, before it does.

Fani 2019

Puri's power-distribution network largely destroyed; prolonged outages in Puri, Bhubaneswar and Cuttack.

Hudhud 2014

Visakhapatnam's power and telecom networks collapsed for days; about ₹21,900 crore in estimated losses.

Amphan 2020

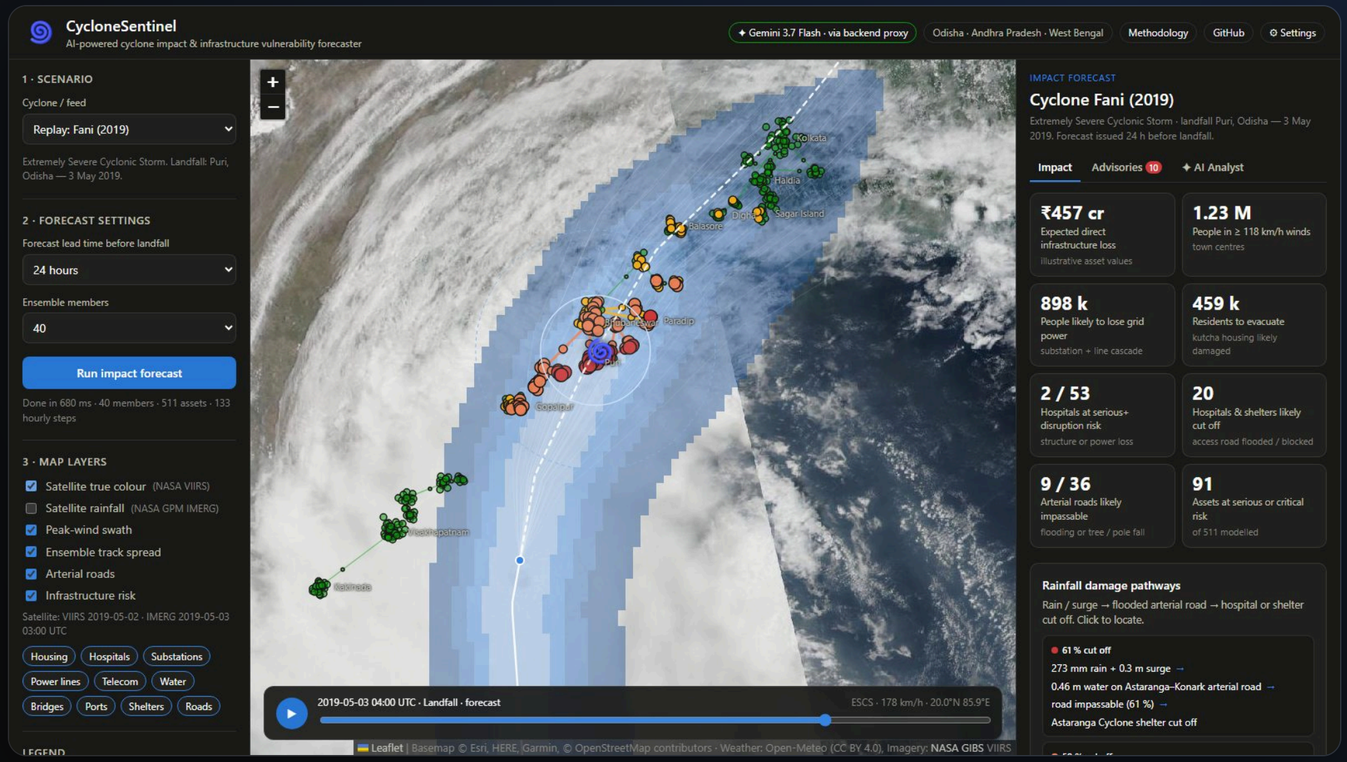
Surge, wind and tree-fall across the Sundarbans and Kolkata; widespread grid and road failure.

- Warnings are issued per **district**; decisions are made per **asset**: this hospital, that substation, this road.
- Failures **cascade**: rain floods an arterial road → a standing hospital is **cut off**; a feeder trips → it goes **dark**.
- Shifting from post-landfall recovery to **pre-landfall action** needs asset-level risk **24–72 h ahead**, and a way to **tell the right authority**.

THE SOLUTION

One platform, four jobs

- **Simulate** wind, **storm surge** and rainfall flooding at 511 assets with a 40-member ensemble
- **Map exposure** of power grids, **arterial roads**, hospitals and **cyclone shelters**
- **Trace rainfall damage pathways**: rain → road flooded → facility cut off
- **Reason with Gemini** over satellite imagery + model output
- **Auto-dispatch** multilingual advisories to district, municipal and state authorities



Fani (2019) replay: real NASA VIIRS imagery under the forecast wind swath

HOW IT WORKS

Satellite + real-time data → physics → Gemini → dispatch

1 · Data feeds

Google Earth Engine: Copernicus DEM, JRC water, WorldPop, IMERG per asset. **NASA VIIRS/IMERG** imagery. **Open-Meteo** real-time 72 h. IMD tracks.

→

2 · Hazard ensemble

40 Monte-Carlo tracks. Holland wind, shelf-amplified **storm surge**, rainfall ponding. Hemisphere-aware.

→

3 · Vulnerability

Fragility curves for 10 asset classes, **power cascade** and **road-access cascade** → P(disruption), P(cut off).

→

4 · Gemini 3.7 Flash

Satellite image + model JSON → situation, pathways, **advisories in Odia / Telugu / Bengali**.

→

5 · Dispatch

IMD colour-coded district advisories → **CAP 1.2**, webhook, e-mail, WhatsApp; **auto-dispatch**.

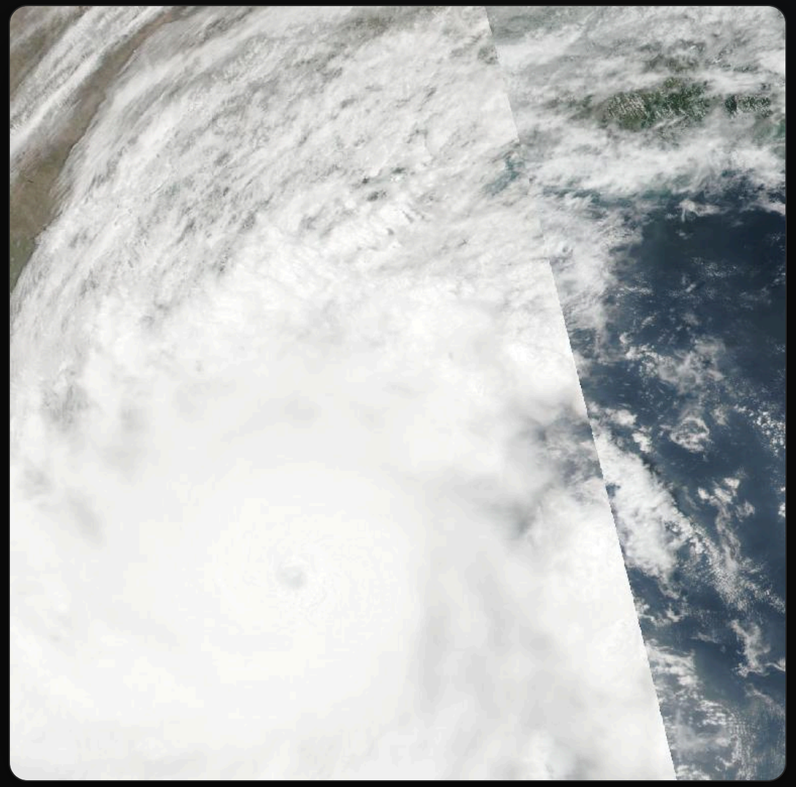
511
assets · 36 arterial roads · 34 towns · 15 districts · 3 states

< 1 s
40 members × 511 assets × 130 hourly steps, in the browser

1 command
to deploy on **Cloud Run**; the static app also runs on GitHub Pages

AI APPROACH

Gemini 3.7 Flash as the duty analyst



Input: NASA VIIRS image, 2 May 2019, sent with the model output

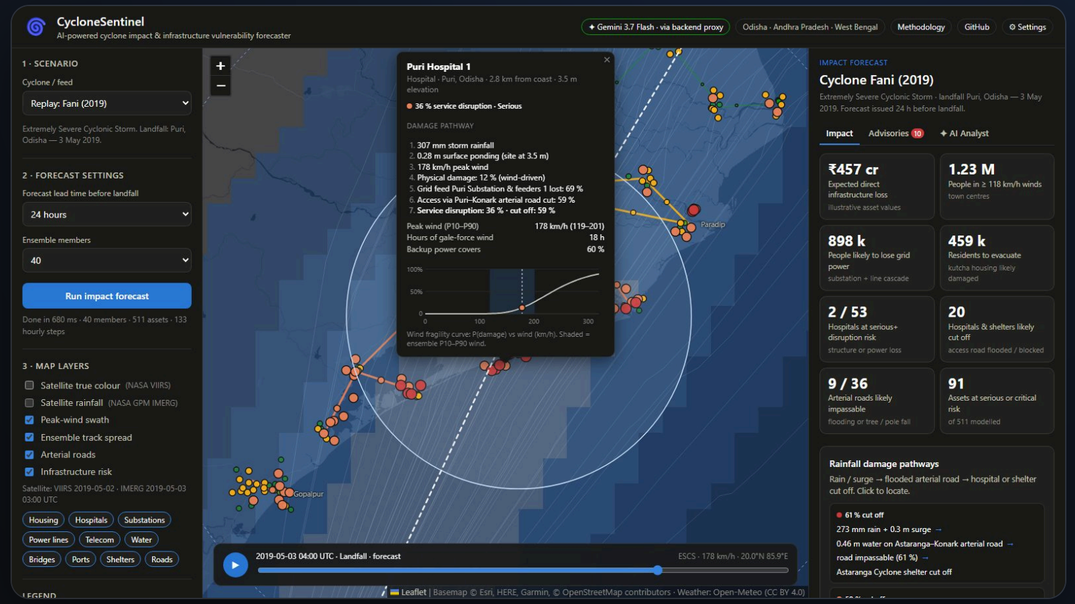
- **Multimodal:** reads the eye, cloud organisation and which coast is under the rain bands
- **Grounded:** gets the model's KPIs, top-risk assets, pathways and advisories as JSON, and is told never to invent numbers
- **Cross-checks** imagery against model hotspots and states where they disagree
- **Structured JSON output:** situation, key risks, rainfall damage pathways with mitigations, confidence, caveats
- **Multilingual:** SMS-ready advisories in English + **Odia, Telugu, Bengali**, attached to the dispatch
- **Reliable:** fallback across Flash models, back-off, key kept server-side on Cloud Run

RAINFALL DAMAGE PATHWAYS

Infrastructure fails as a network

301 mm rain
Fani, Puri
→
0.27 m water
Puri–Konark road
→
Road impassable
59 %
→
Hospital cut off
Puri Hospital 1

- **Road-access cascade:** 20 hospitals and shelters are likely cut off in Fani, even where the buildings survive
- **Power cascade:** line → substation → hospital / water / telecom, net of backup generators
- Each asset carries an **ordered causal chain** the operator can read
- Output is an action: **"Pre-stock 4 hospitals/shelters in Puri for 72 h of isolation"**

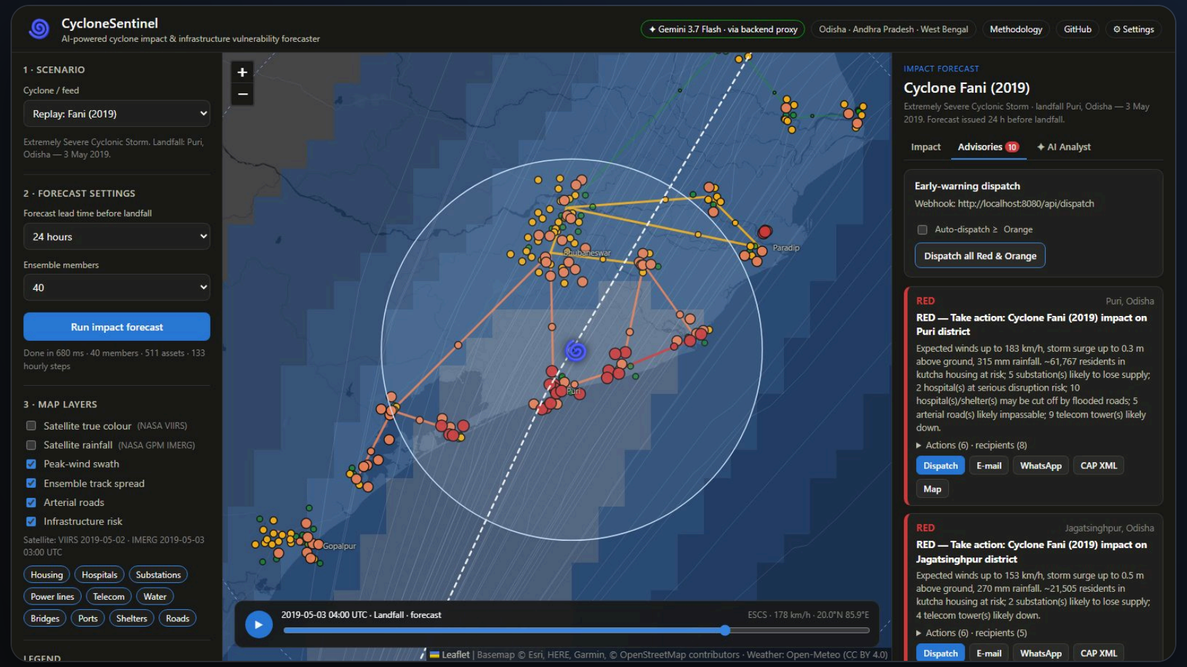


AUTOMATED EARLY WARNING

From forecast to the right desk, automatically

RED **ORANGE** **YELLOW** district advisories on the IMD colour code

- **Routed** to the District Collector & DDMA, the **municipal body** of each affected town, OSDMA / APSDMA / WB DM&CD, the DISCOM and the District Health Officer
- **CAP 1.2 XML**: the format India's **SACHET** platform uses
- Webhook to a Cloud Run relay → control-room inbox, Google Chat, state gateway; plus e-mail and WhatsApp
- **Auto-dispatch** after every run, once per district and level



Fani replay: 10 district advisories, 2 RED (Puri, Jagatsinghpur)

VALIDATION

Hindcast: 4 historical cyclones, forecast issued T−24 h

Cyclone	Actual landfall	Model's worst-hit towns	People likely without power	
Fani (2019)	Puri, Odisha	Puri, Bhubaneswar, Cuttack	9.0 lakh	✓
Amphan (2020)	Sundarbans, West Bengal	Sagar Island, Haldia, Kolkata	7.5 lakh	✓
Phailin (2013)	Gopalpur, Odisha	Berhampur, Gopalpur, Rambha	3.6 lakh	✓
Hudhud (2014)	Visakhapatnam, AP	Visakhapatnam, Vizianagaram, Bheemunipatnam	15.3 lakh	✓

15 automated tests cover the physics (peak wind at the eyewall, right-of-track bias, southern-hemisphere flip), hindcasts, power and road cascades, advisory routing, CAP validity and live mode.

WHO IT SERVES

Built for the 72 hours before landfall

District Collectors & DDMAs

Who to evacuate, to which shelter, which shelters are themselves exposed or cut off.

Municipal bodies (ULBs)

Town-level advisories, pumps and tree-clearing on the arterial roads that will flood.

State SDMAs & EOCs

OSDMA, APSDMA, WB DM&CD: statewide RED/ORANGE picture, CAP feed for SACHET.

Power utilities (DISCOMs)

Substations and line spans likely to fail → stage crews and mobile transformers.

Health departments

Hospitals losing grid power or access → fuel, generators, 72 h stock, transfers.

Citizens, via authorities

Advisories in Odia, Telugu and Bengali, ready for SMS and WhatsApp.

Scale of benefit in the Fani replay alone: **12.3 lakh** people in ≥ 118 km/h winds, **9.0 lakh** likely without power, **4.6 lakh** residents flagged for evacuation.

WHY IT'S DEPLOYABLE

A state could pilot it within weeks

- **Zero install:** a web app; works on any browser in a control room
- **Serverless on Google Cloud:** one `gcloud run deploy`; scales to zero between storms, pay-per-use
- **Standards-based:** CAP 1.2 output for SACHET; IMD colour codes; district and ULB routing
- **Key-safe:** Gemini key stays server-side; the static site shows cached AI output
- **Open source** (MIT) and tested

Pilot plan: one state, 4 weeks

Week 1: load state assets from OSM / Bhuvan / DISCOM GIS; run the GEE pipeline.

Week 2: calibrate fragility on the last 2–3 cyclones' damage reports.

Week 3: connect the dispatch webhook to the state EOC and SACHET test gateway.

Week 4: shadow mode alongside IMD bulletins; human sign-off before any Actual alert.

HOW IT SCALES ACROSS INDIA — AND BRICS

New coast = new data, same engine

- **India:** add Tamil Nadu, Gujarat, Maharashtra and Kerala by adding towns, tracks and assets. The engine, cascades, advisories and CAP are unchanged.
- **Languages:** Gemini writes advisories in each state's language (Tamil, Gujarati, Marathi, Malayalam…)
- **BRICS:** hemisphere-aware physics (tested) serves typhoons in China, the Mozambique Channel (South Africa), the Arabian Gulf (UAE, Iran) and Brazil
- **CAP 1.2** is the WMO/ITU standard used by national alert systems worldwide

Next

- Real asset registers (OSM, Bhuvan, DISCOM GIS) in BigQuery
- Live IMD/JTWC advisories + ECMWF ensembles
- Fragility calibration on Fani, Amphan, Biparjoy, Michaung
- Parametric-insurance triggers from the same probabilities

Try it

blop77.github.io/cyclone-sentinel

github.com/Blop77/cyclone-sentinel · youtu.be/OTejnvAFbVE